Nathan Lapp

Toronto, Ontario • nathanglapp@gmail.com • nathanlapp.xyz

SUMMARY

Gameplay and systems programmer focused on engine architecture, graphics programming, and interactive systems. Experience shipping a Steam title, building a custom game engine, and developing low-level graphics debugging tools alongside VR/MR gameplay systems.

SELECTED PROJECTS

Realm – Custom Game Engine (D)
nathanlapp.xyz/projects/realm.html

- Built a custom game engine in D to explore low-level engine architecture
- Designed and implemented the rendering architecture, plus supporting systems including an Entity Component System, asset loading, and UI

GLTrace – OpenGL Call Tracing & Debugging Tool
nathanlapp.xyz/projects/gltrace.html

- Built a two-component OpenGL tracing system: a low-level function-hooking layer (C/C++) and a Rust-based analysis and tooling pipeline
- Implemented non-invasive capture of runtime graphics calls from running applications through dynamic OpenGL function interception, API hooking, and shared library injection
- Developed the Rust backend that processes and structures captured OpenGL calls into formatted per-frame output

Spider Canyon – 3D Puzzle Platformer (Steam Released)
nathanlapp.xyz/portfolio/spidercanyon/

- Shipped a complete 3D puzzle-platformer on Steam, developed end-to-end from prototype through release
- Designed and implemented custom player movement and traversal systems, including physics-driven mechanics tailored for skill-based navigation and level design
- Built core gameplay systems supporting iterative tuning, level design workflows, and refinement of movement-based challenges

WORK EXPERIENCE

Game Developer Jun 2024 – Present
Invalid Symbol LTD. • Remote

- Architected a modular movement system supporting 20+ abilities for scalable traversal gameplay
- Designed a reusable hierarchical state machine framework used across multiple systems
- Implemented physics-based movement using vector mathematics
- Built JSON-based localization pipeline for text and audio assets
- Implemented VR hand gesture interaction systems for gameplay mechanics
- Led development of a mixed reality game using passthrough occlusion and relighting on Meta Quest

Game Programmer and Consultant Apr 2021 – Sep 2021
Cream Productions • Brampton, Ontario (Remote)

- Developed VR gameplay systems and levels for Meta Quest experiences
- Optimized VR performance using GPU profiling and runtime analysis
- Extended animation pipeline and improved usability and runtime performance in Unity

Research Assistant and Programmer Dec 2020 – Apr 2021
Sheridan Institute of Research and Technology • Brampton, Ontario (Remote)

- Contributed to digital human facial animation pipeline for VR and virtual production
- Built Unity-based framework integrating with facial animation workflows
- Developed tools using image and video processing techniques for experimental pipelines

SKILLS

Core Areas: Engine and gameplay systems; graphics programming and rendering tools; VR/MR interaction systems; networking and real-time systems; API design and systems architecture

Languages: C#, C, C++, Rust, D, Python, GDScript

Engines: Unity, Unreal, Godot

Graphics & Profiling: OpenGL, RenderDoc, Snapdragon Profiler

EDUCATION

B.Sc. Honours, Computer Science – Trent University (2024)
Software Development and Network Engineering – Sheridan College (2021)